

# Pre-Amplifier Based Nanoscale Cross Coupled Dynamic Voltage Comparator for Future Analog to Digital Converters (ADCs): A Reliability Study

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**Abstract**—We reports a high-speed, low-power, dynamic voltage-comparator (DVC) featuring a cross-coupled (CC) pair in the pre-amplifier stage and a strong-arm latch. The suggested modification enhances the pre-amplifier's common-mode (CM) & differential mode gains thereby improving the latch's differential and common-mode input voltages and enabling faster regeneration by 22% compared to the standard comparator during low input differential voltage conditions. The technique further augments the speed while achieving 21% lower energy per conversion delay product (EDP) compared to existing literature. Notably, the delay of the DVC is shown to be immune to the common mode volts. With 16-nm CMOS technology, the comparator measures less than 61.89 pS and 4.529  $\mu$ W power with 0.7 V supply ( $\Delta V_{in}=0.1V$ ). The DVC offered a delay of 27.26 pS and 53.06  $\mu$ W power with 1.1V supply at 300 K. But at 380 K, the DVC offered a delay of 94.64 pS and 4.137  $\mu$ W with 0.7V supply. When  $\Delta V_{in}$  was raised to 0.6V, the DVC offered a delay of 39.34 pS and 4.807  $\mu$ W power with 0.7V supply. Finally at 380 K, when  $\Delta V_{in}$  was raised to 0.6V, the DVC offered a delay of 52.32 pS and 4.327  $\mu$ W power with 0.7V supply.

**Keywords** – ADC, Dynamic voltage, Energy efficient, IoT, VLSI.

## I. INTRODUCTION

The Comparators act as vital interposers that bridge the gap between the physical analog world and the digital sphere, making them necessary factors in a wide array of Analog- To-Digital Converters ( ADCs) [1]. The effectiveness of these ADCs is profoundly affected by the quality and performance of the comparators they use. Accordingly, ongoing exploration in the field of comparator design is devoted to achieving an optimal balance between three vital performance criteria speed, energy effectiveness, and precision. In pursuit of these pretensions, examinations dwells into vital areas similar as minimizing signal detainments, reducing perceptivity to input common-mode voltage ( $V_{i, cm}$ ), and optimizing overall energy consumption. This comprehensive disquisition leads experimenters to estimate a diapason of comparator infrastructures, ranging from the conventional single- tail comparators to the more sophisticated three- stage comparators [2]. Recent advancements have shown the development of different comparator infrastructures, each strictly drafted to surmount distinct challenges. especially, dynamic comparators have surfaced as necessary players in meeting the strict conditions of low- power signal processing operations [3]. However, their susceptibility to noise and common-mode input voltage variations necessitates further enhancement. Simulation-based evaluation has played a key role in identifying sensitive regions within these circuits. For instance, pulsed-laser-induced single-event upSet (SEU)

testing combined with simulation has been successfully used to locate sensitive nodes in dynamic comparator circuits, validating simulation as a valuable tool for robustness analysis and radiation-hardening design efforts [4]

The transition from single- tail to double- tail comparators has proved effective in easing challenges like reflex noise while expanding the  $V_{i,cm}$  range, although this shift has introduced its own set of challenges related to noise and neutralize performance. Nonetheless, the perfection and speed of double-tail comparators constantly wane under conditions of small input discriminational voltages ( $V_{i, id}$ ) and elevated common-mode voltages ( $V_{i,cm}$ ). This performance decline arises from variations in the functional regions of the input transistors, which may shift from achromatism to sub-threshold [5]. The dynamic bias comparator offers an innovative design that concentrated on energy effectiveness, although it does put certain constraints on speed [6]. Engineers are farther exploring energy-effective approaches, similar as the proposed cross-coupled pre-amplifier, which seeks to enhance performance while precisely balancing the trade-off between speed and effectiveness. Despite these advancements, high-reflex noise continues to pose a significant challenge for comparators that incorporate both an amplifier and a latch with delayed activation. In response to these challenges, the foreword of a cross- across-coupled brace in the preamplifier has been linked as a promising avenue for enhancement, serving as the foundation for farther exploration [7]. Relative studies on innovative dynamic latch designs offer perceptivity into key factors, including speed, power consumption, reflex noise, and optimize performance. The optimization and noise characteristics observed in specific latches can serve as critical factors to their use in high-perfection operations, as illustrated through simulation studies [8].

Upon activation, the NMOS input transistors within these latches generally enter the deep triode region, performing in lowered transconductance. This effect becomes pronounced when the drain voltage ( $V_d$ ) ascends towards  $1/ 2V_{DD}$ , causing the gate voltage ( $V_g$ ) to approach the threshold voltage( $V_{thN}$ ). As a consequence, low transconductance hinders the effective coupling of preamplified voltage to the regenerative latch. likewise, the new addition of the cross-coupled brace necessitates the generation of precise timekeeper signals, a factor that significantly influences the speed, neutralize, and noise traits of the DVCs. This improvement is acclimated specifically for latches featuring input transistors that match those of the preamplifier, leading an advanced common-mode gain. Still, incorporating a fresh cascode cross-coupled brace introduces complications, similar

as delayed activation and the necessity for supplementary reset transistors at intermediate bumps. A noteworthy development is the innovative double-tail comparator design, which features a latch along with fresh transistors connected in series with the preamplifier's tail transistor. These fresh transistors are governed by the affair bumps from both the preamplifier and the latch, performing in a comparator that achieves an energy-effective, low-noise result, albeit at the expenditure of comparison speed [9]. To address these limitations, the proposed design introduces an across-coupled brace within the preamplifier, managed by a third current tail for enhanced control. This revision significantly boosts both discriminational and common-mode signal gain, thereby perfecting speed performance [10], especially, the design facilitates the contemporaneous activation of both the cross-coupled brace and the preamplifier, barring the need for fresh reset switches. The proposed preamplifier design maintains harmony with the established Strong-Arm latch, offering a significant enhancement over earlier designs that demanded variations to the traditional Strong-Arm configuration [11]

## II. DYNAMIC COMPARATOR

Dynamic comparators are cleverly designed systems that efficiently differentiate between numbers with low energy use and high speed. This is in contrast to static comparators which check the signals constantly, compared to dynamic comparators which check less often, causing higher power consumption. Dynamic comparators consist of two phases: the preset (also known as reset) and the evaluation phase. During the pre-charging phase, internal nodes in the dynamic comparator are set to a known state. The most noteworthy evaluation phase is when predetermined inputs determine the output as they control the discharge direction, producing a digital pulse that reflects the difference between the signals. Dynamic inline feedbacks that reduce power consumption can increase output pulse speed using regenerative latches or cross coupled inverters. This allows for increased speed, as seen in other devices. These comparators are commonplace in systems that require speed such as ADCs, advanced communication systems, and even memory sense amplifiers. If not, precise timing using time-clocks would lead to charge leakage and metastability problems. Dynamic comparators are highly optimized systems for tasks that need quick response and performance paired with energy efficiency.

## III. CROSS-COUPLED DYNAMIC COMPARATOR

Cross-coupled DVC is a high-speed, low-power, advanced device commonly employed in ADCs and other applications where rapid decision-making is involved. Essentially, the comparator is made up of two strategically cross-coupled interconnection transistors. That implies that the output of one transistor feeds into the input of the other, creating a regenerative feedback loop which has the effect of greatly amplifying an infinitesimal difference between two input signals to huge magnitude. Such a design has the effect of causing the comparator to compare the signals at high speed. One of the characteristics of a cross-coupled dynamic comparator is that it has power consumption efficiency [11]. It is constructed in such a way that it pulls large current loads mainly in switching mode, so it can pull minimal power in the idle condition. Static comparators, which always consume power, are different from this because the dynamic comparator pulls current only at the comparison points of crucial significance when the circuit is stimulated, thus making maximum use of power.

This operational efficiency is especially beneficial for high-speed applications where a decision has to be made immediately without much energy consumption. The cross-coupled topology is also optimized for the speed of the comparison operation, and thus the circuit is very efficient to process high-frequency signals in a very efficient way. Along with the benefits, cross-coupled dynamic comparators also have disadvantages. The decision outputs can be inaccurate due to the effects of noise. The circuit also has metastability problems, meaning it can be in a state where it does not converge to a stable output after some time, resulting in erroneous outputs. Consequently, adequate noise control and increased stability are necessary for the design to ensure that the cross-coupled comparator functions correctly and achieves the goals in practical applications. As noted, Cross-coupled DVCs are a good low-power (LP) and high-performance (HP) dynamic signal processing unit example provided they are tailored for modern electronic systems.

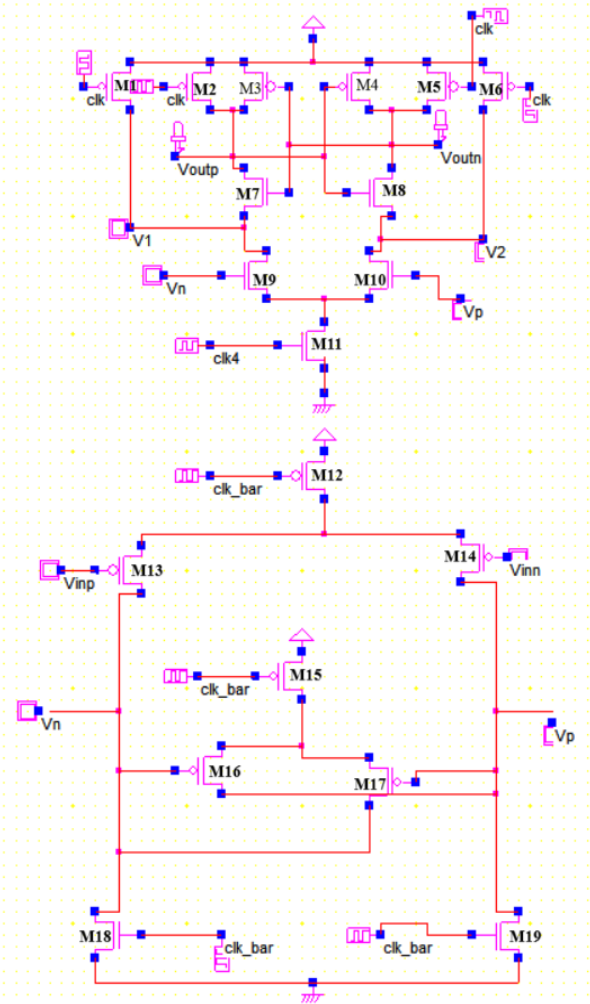


Fig. 1. Circuit diagram of a Crosscoupled comparator [11]

The schematic of the DVC [11] adopted for study is shown in Fig. 1. The pre-amplifier is modified in the proposed design by adding a PMOS cross-coupled brace M16 and M17, which is controlled by the fresh third tail of the comparator M15. The speed performance of the innovative DVC is enhanced by the addition of the cross-coupled brace as the discrimination signal gain of the preamplifier is boosted, and the CM-input for the latch is also increased. The latch has NMOS input transistors; hence, the increase in the CM-input of the latch improves the latch's speed. These advancements hence reduce

the pre-amplifier's detention and the-latch rejuvenescence time. This DVC works in two stages: the reset stage (CLK = "0") and the phase of comparison (CLK = "1"). In the reset stage, the transistors M18 and M19 reset the pre-amplifier affair to GND. also, the switch transistor M2 to M6 resets the latch affair and the intermediate bumps to VDD. The cross-coupled braces M16 and M17 are also turned off by the tail source M15. In the comparison stage, i.e., when CLK is VDD, the tail transistors M12, M15 of the pre-amplifier are actuated, hence cranking the input transistors M13 and M14 as well as the cross-coupled brace M16 and M17. The pre-amplification starts with charging the capacitors (parasitic) at the pre-amplifier nodes Vp and Vn. During modification, the cross-coupled brace transistors M16 and M17 operate in achromatism and boost the discriminational gain as well as the common- mode affair of the pre-amplifier, which can be understood with an illustration.

When  $V_{in} > V_{in}$ , the Vsg of M13 is less compared to that of M14. Hence, Vp charges briskly as compared to Vn, performing in advanced Vsg for M16 in comparison to M17, which increases the charging rate for the Vp knot. thus, the cross-coupled brace increases the discriminational signal, perfecting the latch speed. The fresh cross-coupled brace also increases the affair common-mode voltage of the pre-amplifier as they gives a redundant charging path to the affair bumps.

#### IV. RESULTS OF CROSS-COUPLED DYNAMIC COMPARATOR

##### A. Crosscoupled Comparator with $\Delta V_{in} = 0.1V$

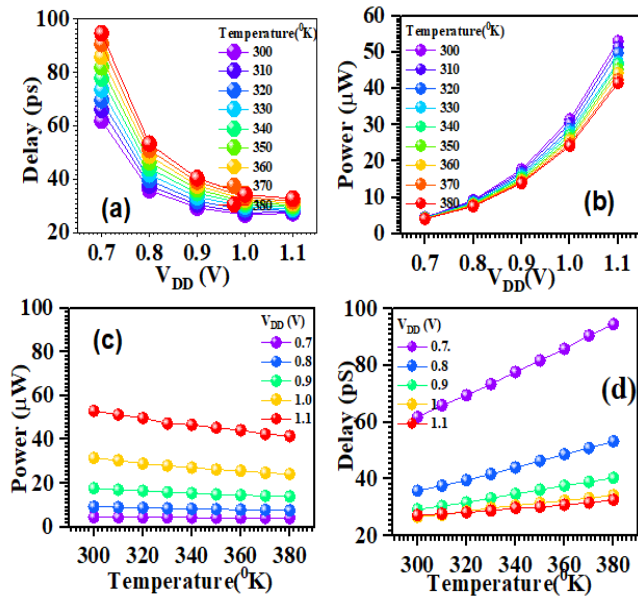


Fig. 2. Power and delay of a Crosscoupled comparator at  $\Delta V_{in}$  is 0.1V

The Cross-coupled comparator operated with  $V_{DD}$  ranges from 0.7V to 1.1V with a  $\Delta V_{in}$  of 0.1V to 0.6V. The delay of 61.89 pS is illustrated at 0.7V and 300K, and the delay gradually increases to 91.64 pS at 380K the delay gradually varies with supply at different temperatures as shown in Fig.2(a). The supply was increased to 1.1V, then the delay of 27.26 pS is illustrated at 300K and gradually increases to 32.69 pS at 380K by varying temperatures from 300K to 380K respectively. The power consumed by the circuit at 0.7V and 300K is demonstrated as 4.529  $\mu$ W, and the power gradually decreased to 4.137  $\mu$ W at (380) Kelvin at 1.1V and power drastically rises to 53.06  $\mu$ W illustrated at 300K and

drastically changing to 41.58  $\mu$ W at 380K. As demonstrated in Fig. 2(c), the temperature vs power at different Supplys ( $V_{DD}$ ). At a temperature of 300K when the  $V_{DD}$  is set to 0.7V, the delay is illustrated as 61.89 pS, and by gradually increasing the supply ( $V_{DD}$ ) up to 1.1V at a  $\Delta V_{in}$  of 0.1V, the delay is demonstrated as 27.26 pS i.e, the delay drastically falls. The delay at 380K and 0.7V is demonstrated as 94.64pS and as the  $V_{DD}$  gradually increases to 1.1 V then the delay respectively decreases as Fig. 2(c) and The power consumed by the DVC is demonstrated as 4.529  $\mu$ W at a temperature of 300K at 0.7V and it drastically increases to 53.06  $\mu$ W at same temperature when supply increases to 1.1 V. The power consumed by the DVC varied from 4.137 to 41.58  $\mu$ W when the temperature was at 380K, and the supply was increased from 0.7V to 1.1V.

##### B. Crosscoupled Comparator at $\Delta V_{in} = 0.2V$

As shown in Fig. 3, the supply is shifted from 0.7V to 1.1V with a  $\Delta V_{in}$  of 0.1V to 0.6V at different temperatures. The circuit delay is illustrated as 50.78 pS at a supply of 0.7V and a temperature of 300K when  $\Delta V_{in}$  is 0.2V. Then, the delay gradually increases to 75.19 pS at 380K at the same supply. At 1.1V, the delay of the circuit is demonstrated as 25.69 pS at 300K, the delay gradually rises to 30.66 pS at a temperature of 380K observed at different temperatures varying from 300 Kelvin to 380 Kelvin, respectively.

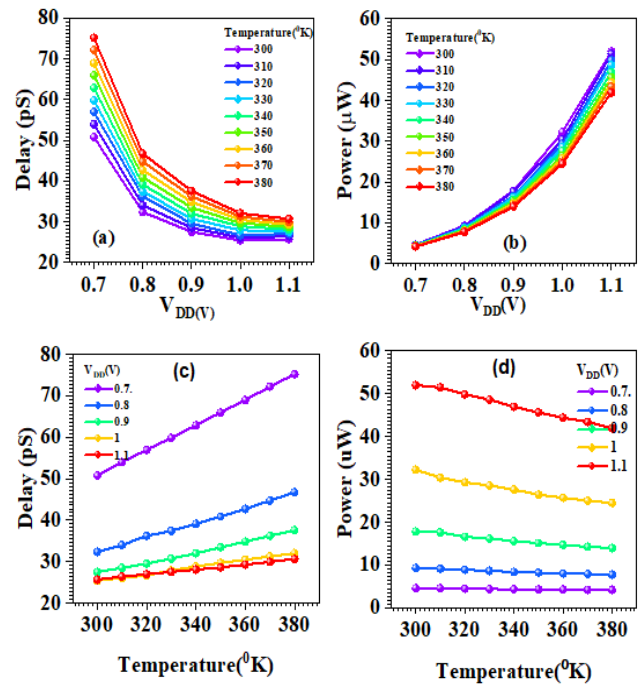


Fig. 3. Power and delay of a Crosscoupled comparator at  $\Delta V_{in}$  is 0.2V

Power consumed by the DVC is illustrated as 4.625  $\mu$ W at 0.7V and 300K when  $\Delta V_{in}$  is 0.2V the consumption dropping to 4.181  $\mu$ W at 380K and at 1.1V, it jumps to 51.97  $\mu$ W, decreasing slightly to 41.92  $\mu$ W at 380K Overall, as circuit delay decreases, consumption of power increases at different temperatures ranging from 300K to 380K as plotted in Fig. 3(a) & Fig.3(b), respectively. In Fig. 3(c), the delay vs temperature of the circuit is demonstrated at different supplys. The delay is illustrated as 50.78pS at a supply of 0.7V at a temperature of 300K, and when the supply is increased to 1.1V, the delay gradually decreases to 25.69pS. When the temperature is increased to 380K then the delay is demonstrated as 75.19 pS, and it drastically reduces to 30.66

pS. The power consumed by the DVC across different temperatures varying supply is illustrated as  $4.625\mu\text{W}$  at  $0.7\text{V}$ ,  $52\mu\text{W}$  at  $1.1\text{V}$  at  $\Delta V_{in}$  is  $0.2\text{V}$ , and when the temperature was increased to  $380\text{K}$ , the power is demonstrated as  $4.181\mu\text{W}$  and drastically increases to  $41.92\mu\text{W}$ .

### C. Crosscoupled Comparator at $\Delta V_{in} = 0.3\text{V}$

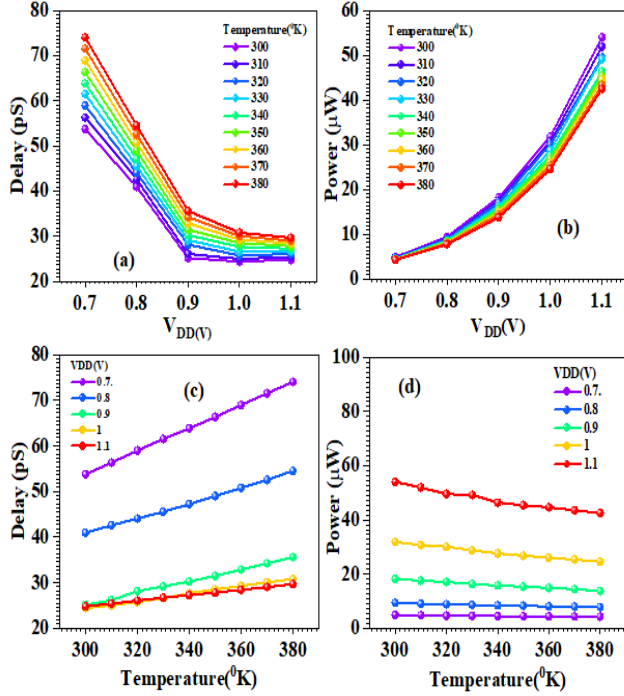


Fig. 4. Power and delay of a cross-coupled comparator at  $\Delta V_{in}$  is  $0.3\text{V}$

In Fig. 4, the circuit's delay is illustrated by varying the power supply from  $0.7\text{V}$  to  $1.1\text{V}$  with different temperatures, and the  $\Delta V_{in}$  is varied from  $0.1\text{V}$  to  $0.6\text{V}$ . The delay of the circuit at a supply of  $0.7\text{V}$  and at a temperature of  $300\text{K}$  with  $\Delta V_{in}$  of  $0.3\text{V}$  is  $53.78\text{pS}$ , and when the temperature is increased to  $380\text{K}$ , the delay of the circuit gradually increases to  $74.04\text{pS}$ . The delay is illustrated as  $24.79\text{pS}$  at a power supply of  $1.1\text{V}$ , and it gradually increases to  $29.69\text{pS}$  as shown in Fig. 4(a). The power consumed by the DVC is demonstrated as  $4.878\mu\text{W}$  at a supply of  $0.7\text{V}$  and a temperature of  $300\text{K}$ , when the temperature is increased to  $380\text{K}$ , the power gradually decreases slightly to  $4.253\mu\text{W}$  and when the power supply of the circuit is increased from  $0.7\text{V}$  to  $1.1\text{V}$ , then the power is illustrated as  $54.02\mu\text{W}$  and it gradually decreases to  $42.53\mu\text{W}$  when the temperature is varying from  $300\text{K}$  to  $380\text{K}$  respectively as shown in Fig 4(b). In Fig 4(c), the circuit delay has been decreased from  $53.78\text{pS}$  to  $24.79\text{pS}$  at a temperature of  $300\text{K}$  and a  $\Delta V_{in}$  of  $0.3\text{V}$ , and when the temperature is varying from  $300\text{K}$  to  $380\text{K}$ , the delay of that circuit has been drastically increased to  $74.04$  at a supply of  $0.7\text{V}$  and it gradually falls to  $29.69\text{pS}$  as shown in Fig. 4(c). The power of the circuit is illustrated as  $4.878\mu\text{W}$  at a supply of  $0.7\text{V}$  at a temperature of  $300\text{K}$  and it drastically increased to  $54\mu\text{W}$  and conversely when the supply is increased to  $1.1\text{V}$ , and at  $380\text{K}$  the power is demonstrated as  $4.253\mu\text{W}$  and increased to  $42.53\mu\text{W}$  when the supply is adjusted to  $1.1\text{V}$  as shown in Fig.4(d). The DVC exhibits excellent reliability even at high temperatures in terms of power and speed.

### D. Crosscoupled Comparator at $\Delta V_{in} = 0.4\text{V}$

The cross-coupled DVC operated with various supply varying from  $0.7\text{V}$  to  $1.1\text{V}$  at different temperatures varying from  $300\text{K}$  to  $380\text{K}$  at different reference voltages. The delay of the circuits is illustrated as  $44.56\text{pS}$  at a supply of  $0.7\text{V}$  and a temperature of  $300\text{K}$  at  $\Delta V_{in}$  is set to  $0.4\text{V}$  and when the temperature is varying from  $300\text{K}$  to  $380\text{K}$ , the delay has been gradually increased to  $60.64\text{pS}$ . When the supply is increased to  $1.1\text{V}$  the delay of the circuit is demonstrated as  $24.32\text{pS}$  conversely when temperature increased to  $380\text{K}$  the delay is illustrated as  $29.07\text{pS}$  as shown in Fig.5(a). The power consumed by the DVC is illustrated as  $4.895\mu\text{W}$  at a temperature of  $300\text{K}$  and  $0.7\text{V}$  and when temperature is varying to  $380\text{K}$ , the power consumed by the circuit is illustrated as  $4.281\mu\text{W}$ . Conversely when supply is increased to  $1.1\text{V}$ , the power of the DVC is illustrated as  $54.56\mu\text{W}$  at  $300\text{K}$  and it drastically falls to  $42.67\mu\text{W}$  at  $380\text{K}$  as shown in Fig.5(b).

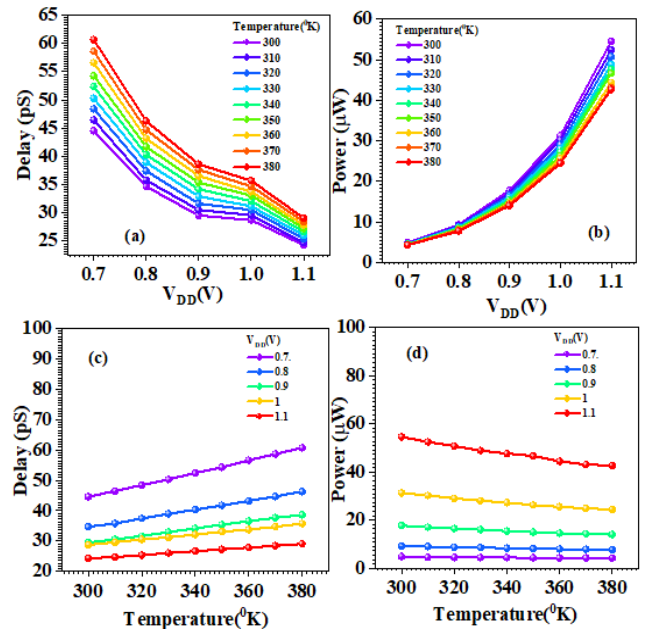


Fig. 5. Power and delay of a cross-coupled comparator at  $\Delta V_{in}$  is  $0.4\text{V}$

In Fig.5(c), the delay of the circuit is illustrated as  $44.56\text{pS}$  when operated at a voltage of  $0.7\text{V}$  at a temperature of  $300\text{K}$ , and it drastically falls to  $24.32\text{pS}$  at  $1.1\text{V}$ . Conversely, when the temperature is rising from  $380\text{K}$  then the delay is demonstrated as  $60.64\text{pS}$  and it drastically decreased to  $29.07\text{pS}$  as shown in Fig.5(c). The power vs temperature is demonstrated in Fig.5(d), the graph illustrates that when the circuit is operated at a supply of  $0.7\text{V}$ , the power consumed is demonstrated as  $4.895\mu\text{W}$  and drastically increased to  $54.56\mu\text{W}$  at  $1.1\text{V}$  at  $300\text{K}$  and when the temperature is increasing from  $300\text{K}$  to  $380\text{K}$ , the power consumption decreased to  $4.281\mu\text{W}$  at  $0.7\text{V}$  and it drastically increased to  $42.67\mu\text{W}$  when supply is varying from  $0.7\text{V}$  to  $1.1\text{V}$ .

### E. Crosscoupled Comparator at $\Delta V_{in} = 0.5\text{V}$

In Fig. 6, the circuit's supply ranges from  $0.7\text{V}$  to  $1.1\text{V}$ , with a reference voltage of  $0.1\text{V}$  to  $0.6\text{V}$ . At a power supply of  $0.7\text{V}$  at  $300\text{K}$ , the delay of the circuit is demonstrated as  $40.92\text{pS}$ , and when temperature varies from  $300\text{K}$  to  $380\text{K}$ , the delay drastically increases to  $55.23\text{pS}$ . Conversely, at a supply of  $1.1\text{V}$ , the delay illustrated as  $26.76\text{pS}$ , and it slightly the delay of the circuit increases to  $32.49\text{pS}$  as shown in



Fig.6(a). The power consumed by the circuit is demonstrated as 4.779  $\mu\text{W}$  at a supply of 0.7V at 300K, and it slightly decreases to 4.297  $\mu\text{W}$  at 380K. When the supply varies from 0.7V to 1.1V, the power is illustrated as 52  $\mu\text{W}$  and gradually falls to 42  $\mu\text{W}$  as shown in Fig.6(b).

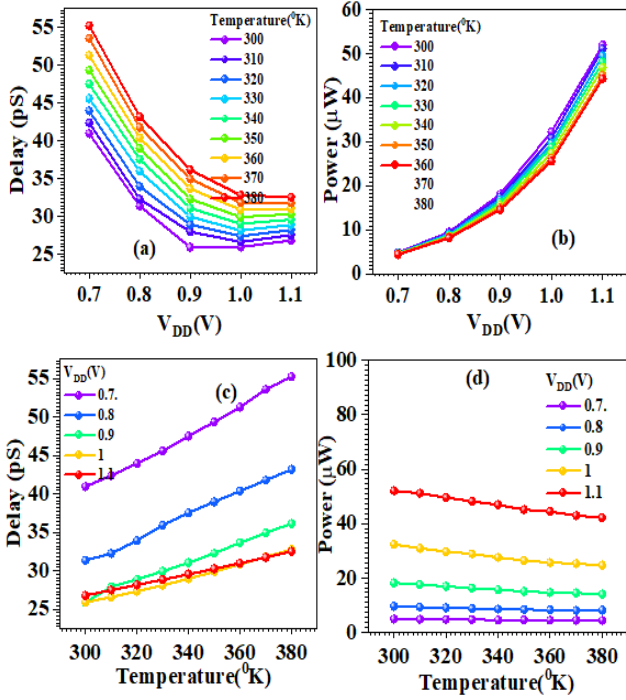


Fig. 6. Power and delay of a cross-coupled comparator at  $\Delta V_{in}$  is 0.5V

The delay illustrated at 300K, when the circuit is operated with different supplies varying from 0.7V to 1.1V at  $\Delta V_{in}$  is 0.5V, and when the supply is set to 0.7V, the delay is demonstrated as 40.92 pS and by varying power supply to 1.1V the delay drastically falls to 26.76 pS and when the temperature is increased to 380K the delay of the circuit is decreasing from 55.23 to 32.49 pS when supply is varying from 0.7V to 1.1V as shown in Fig. 6(c). Conversely, the power consumed by the demonstrated as 4.779  $\mu\text{W}$  at a temperature of 300K and supply of 0.7V and drastically increases to 52  $\mu\text{W}$  at 1.1V and a temperature of 380K the power consumption is illustrated as 4.297  $\mu\text{W}$  and it increased to 42.09  $\mu\text{W}$  at power supply of 1.1V as shown in Fig. 6(d).

#### F. Crosscoupled Comparator at $\Delta V_{in} = 0.6V$

The cross-coupled comparator operated with supplies varying from 0.7V to 1.1V at different  $\Delta V_{in}$  ranges from 0.1V to 0.6V. The delay of the circuit illustrated when the supply is 0.7V at 300K is 39.34 pS and when the temperature is changing to 380K, the delay is demonstrated as 52.32 pS and when the supply is increasing from 0.7V to 1.1V the delay is varying from 25.23 pS to 30.5 pS, when temperature is varying from 300K to 380K respectively as shown in Fig.7(a). The power consumed by the DVC illustrated as 4.8  $\mu\text{W}$  at a supply of 0.7V at 300K, it slightly decreases to 4.327  $\mu\text{W}$  when temperature is varying from 300K to 380K and when the supply is varying from 0.7V to 1.1V the power consumed by the DVC is demonstrated as 53.5  $\mu\text{W}$  at 300K and when the temperature is increasing to 380K the power consumption slightly falls to 41.75  $\mu\text{W}$  as shown in Fig.7(b).

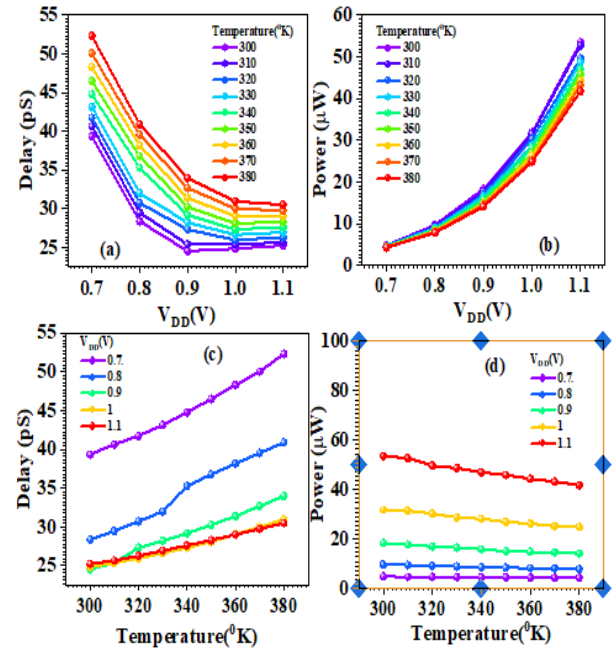


Fig. 7. Power and delay of a cross-coupled comparator at  $\Delta V_{in}$  is 0.6V

#### G. Temperature Effects

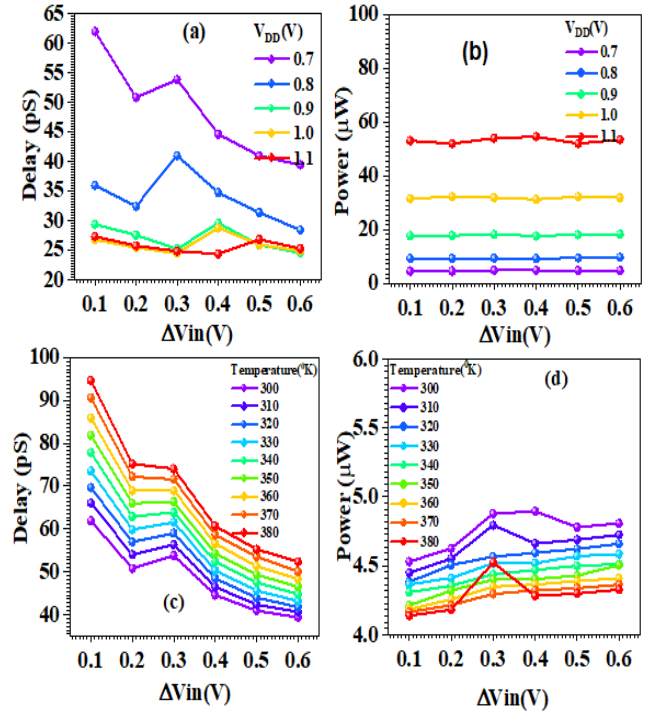


Fig. 8. Delay and power at different temperatures and supplies.

In the above Figure, the delay vs  $\Delta V_{in}$  graph operated at different supplies varying from 0.7V to 1.1V at 300K is plotted. When the  $\Delta V_{in}$  is varying from 0.1 to 0.6 at a supply of 0.7V, the delay is illustrated as 61.89pS and decreased to 27.26pS when supply is varies from 0.7V to 1.1V and at  $\Delta V_{in}$  is 0.6V the delay is illustrated as 39.34pS and it decreases slightly to 25.23pS with varying supplies respectively as shown in Fig.8(a). The power consumed by the DVC is illustrated as 4.529  $\mu\text{W}$  at 0.7V, and it drastically increased to 53  $\mu\text{W}$  by varying the supply from 0.7V to 1.1V. The delay of the cross-coupled comparator is demonstrated as 62.89pS, and it varies to 94.64pS with varying the temperature at  $\Delta V_{in}$  is

0.1V and at  $\Delta V_{in}$  is 0.6V the delay varies from 39.34pS to 52.32pS when temperature is varying from 300K to 380K as shown in Fig.8(c). The power consumed by the cross-coupled comparator circuit at  $\Delta V_{in}$  is 0.1V is demonstrated as 4.529  $\mu$ W at 300K and decreased to 4.137  $\mu$ W at 380K and at  $\Delta V_{in}$  is 0.6V the power consumption varied from 4.807  $\mu$ W to 4.327  $\mu$ W when temperature is changing from 300K to 380K as shown in Fig.8(d).

#### IV. CONCLUSION

In summary, Cross-coupled comparators are widely used in high-speed and low-power electronic systems for their fast decision-making capabilities, and these comparators are highly sensitive to a minimal voltage reference. In this work, we studied the reliability of a cross-coupled comparator with 21 transistors consisting of a modified pre-amplifier and a strongarm latch. and simulated for the supply varying from 0.7V to 1.1V at temperatures varying from 300K to 380K at different voltage references from 0.1V to 0.6V, the results are very precise for various reference voltages and temperatures at different voltages. This innovative DVC can be further enhanced in future by realizing it using emerging transistors such as 2D-FETs (graphene FETs, CNTFETs) or nanosheet (NS)-FETs or Forksheet (FS)-FETs.

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